Tim Sippel, PhD

Senior Aquatic and Marine Scientist

I am a senior environmental scientist, specialising in aquatic and marine ecology. My 25+ years of experience includes the last 10 years working in the Pacific Northwest on salmonid science and population management with state, Federal, and tribal co-managers. The range of projects I have worked on include dam relicencing, dam removal, population dynamics including survival analysis and abundance estimation, spawning habitat restoration, and ESA section 7 consultations in Washington, Oregon, California and Alaska.

Contact

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Work History (since completing PhD)

Stantec, Senior Environmental Scientist - Fish Ecology SME: Jul 2024 - Present

Position: Senior scientific and analytical support for regulatory compliance projects including hydroelectric dam relicensing and removals, Biological Assessments, ESA Section 7 consultations, habitat restoration and mitigation, onshore and offshore renewable energy, site remediation, water quality, water management and conveyance, and fishery management. Data analytic and statistical support for diverse projects.

Accomplishments: Senior author developing manuscript on spring Chinook spawning habitat carrying capacity, and thought leadership and technical contributions to California Centeral Valley Improvement Project and National Academies of Science committees and presenting at scientific meetings. Member of North America Quantitative Ecology team, statistical and analytical support for diverse projects. Contributed to procurement of new projects and mentored junior staff.

Worley, Senior Environmental Scientist - Fish Ecology SME: Jun 2022 - Jun 2024

Position: Senior scientific and analytical support for regulatory compliance projects including offshore renewable energy, climate change risk assessment, salmon population assessment, assessment of fishing within global marine protected areas

Accomplishments: Procurred contracts with WDFW to develope and document new analytical tools for coho salmon population and fish impact assessment, and with The Nature Conservancy to look at recreational fisheries in global marine protected areas. Analytical support for a variety of projects, including development of a climate change risk assessment toolbox. Presented at scientific conferences and mentored junior staff.

Washington Department of Fish and Wildlife, Fishery Scientist: Aug 2017 - Aug 2022

Position: Monitoring, forecasting and reporting on salmon and steelhead population abundance for fishery compliance with the Endangered Species Act and US vs Oregon Management Agreement. Close coordination with other state, Federal and tribal fishery managers. WDFW lead representative to US vs Oregon Technical Advisory Committee. Worked closely with tribal co-managers, including CRITFC on mainstem Columbia River salmon and steelhead management, and Conferaded Tribes of the Colville Nation.

Accomplishments: Initiated and lead data modernization and analysis projects to improve transparency, accuracy and efficiency. Secured funding for, and lead development of a database for Columbia River salmon population assessment and regulatory compliance reporting. Contributed new thought leadership to technical issues on data management and analysis with direct application to agency obligations and priorities.

Ocean Associates (NOAA Contractor), Research Biologist: Sep 2011 - Jul 2017

Position: Conducted fishery stock assessments as a member of international and domestic scientific working groups. Served as NMFS representative to the Pacific Fishery Management Council's Highly Migratory Species Management Team advising on fishery management policy.

Accomplishments: Made key scientific contributions to both domestic and international fishery management as stock assessment scientist and scientific advisor. Senior authorship of two peer-reviewed scientific publications.

University of Hawaii, Post-doctoral Scientist: Mar 2010 - Aug 2011

Position: Develop models to estimate behavior of fish carrying satellite transmitters, and contribute to development and deployment of databases to manage satellite tagging data.

Accomplishments: Developed tools for analyses of animal movement ecology, including R codebases and Access databases for large electronic tagging datasets. Co-authored a peer-reviewed book chapter on juvenile white sharks.

Technical Skills

- **R programming**: tidyverse, time-series, machine learning, visualization
- **GIS**: R packages: sf, terra, leaflet; QGIS
- **Quarto**: Parameterized reporting
- **Data management**: SQL, GitHub, build and manage relational databases,
- **Specialized Software**: Stock Synthesis, Positron, RStudio
- **SCUBA**: AAUS Scientific Diver, NAUI Nitrox Certified

Project Experience

Details →

Remediation (Oregon); **Fish passage** (Oregon, Washington); **Water quality** (Ontario, Canada); **Aquatic habitat improvement** (Alaska); **Dam relicensing** (Oregon); **Marine and estuarine habitat monitoring** (California); **Dam removal** (Washington); **Habitat supplementation** (California); **Marine fiber optic cable** (Pacific Ocean basin); **Aquatic habitat loss mitigation** (Peru); **Water conveyance** (US West coast); **Offshore wind feasibility** (US West Coast, US Pacific Islands, Australia); **Marine Protected Areas** (Global); **Salmon population assessment** (Oregon); **Fishery protection plan** (New Jersey); **Climate change risk assessment** (US & International)

Professional Service

California Central Valley Project Improvement Act, Science Integration Team: 2025 - Present

National Academies of Sciences, Engineering, and Medicine, Editorial Committee for the US National Assessment by the Nature Record: 2026

Ministry for Primary Industries - Fisheries New Zealand, Member of Working Groups for Highly Migratory Species and Deepwater Species: 2026 - Present

National Academies of Sciences, Engineering, and Medicine, Standing Committee on Offshore Wind and Fisheries Member: 2025 (committee dissolved)

US vs Oregon Policy Committee, Technical Advisory Committee Member: 2017 - 2022

Pacific Fisheries Management Council, Highly Migratory Species Management Team: 2012 - 2017

International Scientific Committee for Tuna and Tuna-like Species, Stock Assessment Analyst: 2011 - 2017

Education

PhD - Marine Ecology, University of Auckland, New Zealand

MSc - Marine Ecology, Massey University, New Zealand

B.S. - Biology, Kansas State University, Manhattan, KS

Professional Memberships

American Fisheries Society: 2023 - Present

Selected Publications

See all publications →

Peer-reviewed publications

Sippel, T., Bernazzani, P., … (2026). Uncertainty and bias in estimating salmon spawning habitat carrying capacity: A review of best available science, a management case study, and lessons learned. *In prep*.

Francis, M., Lyon, W., … **Sippel, T.** (2023). Post-release survival of shortfin mako sharks released from pelagic tuna longlines in the Pacific Ocean. *Aquatic Conservation: Marine and Freshwater Ecosystems* 33: 366–378.

Sippel, T., Lee, H., … (2017). Searching for M: Is there more information about natural mortality in stock assessments than we realize? *Fisheries Research* 192: 135–140.

Maxwell, S., Hazen, E., … **Sippel, T.** (2015). Dynamic ocean management: Defining and conceptualizing real-time management of the ocean. *Marine Policy* 58: 42–50.

Sippel, T., Paige Eveson, J., … (2015). Using movement data from electronic tags in fisheries stock assessment: A review of models, technology and experimental design. *Fisheries Research* 163: 152–160.

Evans, K., Abascal, F., … **Sippel, T.** (2014). The horizontal and vertical dynamics of swordfish in the South Pacific Ocean. *Journal of Experimental Marine Biology and Ecology* 450: 55–67.

Weng, K., O'Sullivan, J., … **Sippel, T.** (2012). Back to the wild: Release of juvenile great white sharks from the Monterey Bay Aquarium. *Global Perspectives on the Biology and Life History of the Great White Shark (Carcharodon carcharias)*: 419–446.

Sippel, T., Holdsworth, J., … (2011). Investigating Behaviour and Population Dynamics of Striped Marlin (Kajikia audax) from the Southwest Pacific Ocean with Satellite Tags. *PLoS ONE* 6: e21087.

Holdsworth, J., **Sippel, T.** & Block, B. (2009). Near real time satellite tracking of striped marlin (Kajikia audax) movements in the Pacific Ocean. *Marine Biology* 156: 505–514.

LePort, A., **Sippel, T.** & Montgomery, J. (2008). Observations of mesoscale movements in the short-tailed stingray (Dasyatis brevicaudata) from New Zealand using a novel PSAT attachment method. *Journal of Experimental Marine Biology and Ecology* 359: 110–117.

Sippel, T., Davie, P., … (2007). Striped marlin (Tetrapturus audax) movements and habitat utilization during a summer and autumn in the Southwest Pacific Ocean. *Fisheries Oceanography* 16: 459–472.

Reports

National Academies of Sciences, Engineering, and Medicine (2026). *Offshore Renewable Energy Development on the West Coast: Understanding Effects on Shipping, Fisheries, and Maritime Activities*. National Academies Press.

National Academies of Sciences, Engineering, and Medicine (2026). *Review of the Draft U.S. National Assessment by The Nature Record*. The National Academies Press.

Informal Offshore Wind Energy Group (2024). *Oregon Floating Offshore Wind Energy Roadmap with Exit Ramps*. Oregon Consensus.

Cox, B. & **Sippel, T.** (2020). Pound Nets and PD17 2017 Survival Analysis. *Washington Department of Fish and Wildlife*: 12.

Stohs, S. & **Sippel, T.** (2017). Analysis of Increasing the Required VMS Ping Rate for the California Drift Gillnet Fishery. *NOAA-NMFS*: 18.

Kai, M., Carvalho, F., … **Sippel, T.** (2017). Stock assessment for the north Pacific Blue shark (Prionace glauca) using Bayesian State-space Surplus Production Model. *NOAA*: 43.

Carvalho, F., Kai, M., … **Sippel, T.** (2017). Stock Assessment of Blue Shark in the North Pacific Ocean Using Stock Synthesis. *NOAA*: 52.

Carvalho, F. & **Sippel, T.** (2016). Direct Estimates of Gear Selectivity for the North Pacific Blue Shark Using Catch-at-Length Data: Implications for stock assessment. *NOAA - Pacific Islands Fisheries Science Center*: 10.

Sippel, T., Ohshimo, S., … (2014). Spatial and temporal patterns in size and sex of shortfin mako sharks from US and Japanese commercial fisheries: a synthesis to guide future research. *NOAA*: 36.

Holdsworth, J. & **Sippel, T.** (2008). A summary of New Zealand southern bluefin tuna electronic tagging results. *New Zealand Ministry of Fisheries*: 25–25.